THREE CONJECTURES ON ALTERNATING PLANE GRAPHS

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Abstract. Althöfer, Haugland, Scherer, Schneider and Van Cleemput introduced *alternating plane graphs* and closed their paper with four open problems. We settle two and advance a third.

Conjecture 10.1, that no alternating plane graph has exactly two distinct vertex degrees or exactly two distinct face sizes, is proved by one per-edge inequality applied twice. With face size counted by boundary-walk length throughout, the same conclusion is then proved again when face alternation is required only between distinct adjacent faces, so that bridges are admitted.

Conjecture 10.2, that $(3, 4, 5)$-alternating plane graphs exist for every $n \geq 20$, is settled by verified certificates at the 26 orders that were open, together with a periodic capping lemma reaching order 48 and every $n \geq 50$; these close every $n \geq 46$ independently of the original paper.

For Conjecture 10.3 we give 3-connected witnesses at 54 orders; its infinite tail rests on an induction we do not establish. The fourth problem is untouched.

1. Introduction

Althöfer, Haugland, Scherer, Schneider and Van Cleemput [1] introduced alternating plane graphs: plane graphs in which adjacent vertices have different degrees and adjacent faces have different sizes. The condition is severe. It forces a global rigidity that makes even existence questions hard, and the paper closes ([1], §10, p. 362) with four open problems. Two are settled here, and the third is advanced but not closed.

Results.

- Conjecture 10.1 is proved (Theorem 4.1), and again for the bridge-relaxed definition of §5.
- Conjecture 10.2 is settled, by certificates together with a proved construction (Theorem 6.1, Corollary 6.3).
- Conjecture 10.3 is *not* settled here. Theorem 7.1 gives 3-connected witnesses at 54 orders; the tail above 56 is left open (Remark 7.3).
- The asymptotic degree distribution is left open (§8).

Disclosure of AI use. Large language models were used throughout this work and the extent is material. By task:

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Key words and phrases. alternating plane graph, plane graph, rotation system, computer-assisted proof.

- *Software.* The four decision procedures, the search programs that found the certificates, and the supporting tooling were written with AI assistance.
- *Mathematics.* The step in §5 that drops the (C2) hypothesis, extending Theorem 4.1 to bridge-relaxed graphs, was proposed by an AI reviewer. The author re-derived it independently, and every bound it uses is confirmed by exhaustive search over the relevant degree and face-size ranges in exact rational arithmetic.
- *Adversarial review.* AI review found errors the author had missed, including three claims stated in earlier drafts that were false and were withdrawn. These are recorded at the files that made them rather than deleted.
- *Drafting.* The manuscript was drafted with AI assistance. The author has verified its mathematical content and is answerable for it and for this disclosure; the prose is not claimed to be the author's own sentence by sentence.

No AI system is an author of this paper, and responsibility for every claim in it rests with the named author. The artifact [2] names the systems and versions used, and states which of its own conclusions were taken on trust.

On the use of computation. Two of the three results assert the existence of specific graphs, and those graphs are large: the certificates here run to 110 vertices. We state them as rotation systems, which determine a plane map up to homeomorphism, and we derive every claimed property — degrees, faces, face sizes, bridges, connectivity — rather than storing it. Nothing in the artifact records a claim that a graph *is* an alternating plane graph; separately written decision procedures recompute that on every run — three for the $(3, 4, 5)$ class, and a fourth for the general class of Definition 2.1, which Conjecture 10.3 needs at order 19.

The proofs in §4 and §6 are ordinary mathematics and can be read without a computer. Computation enters in two places: verifying the certificates, and discharging two finite hypotheses of the capping lemma, both stated explicitly in §6 so that a reader can see precisely what is being taken from a machine.

2. Preliminaries

We follow [1] throughout.

Definition 2.1 ([1], Definition 2.1)**.** An *alternating plane graph* is a simple connected plane graph in which

(a) every vertex has degree at least 3;
(b) every face, the unbounded one included, has size at least 3;
(c) adjacent vertices have different degrees;
(d) adjacent faces have different sizes.

Definition 2.2 ([1], Definitions 3.1 and 3.4)**.** A $(3, 4, 5)$-alternating plane graph is one in which every vertex degree and every face size lies in $\{3, 4, 5\}$. An X, Y*-alternating plane graph* is one with exactly X distinct vertex degrees and exactly Y distinct face sizes.

Convention 2.3 (face size)**.** The *size* of a face is the length of its boundary walk counted with multiplicity, that is, the number of edge-side incidences. It is not the number of distinct vertices on the face. The two differ exactly when the walk repeats a vertex, which is a property of one face and not of the graph. In particular a bridge contributes 2 to the size of the single face carrying it.

[1] does not separately define "size". This is nonetheless the reading its own arguments require, in three independent places. Equations (3.1) and (9.2) both assert $\sum_s s\, f_s = 2e$, which counts edge-side incidences. The proof of Lemma 9.2 states that, the graph being bipartite, $f_j = 0$ for every odd j — valid for boundary-walk length, and *not* in general for the number of distinct boundary vertices, since a bipartite facial walk through a cut vertex can meet an odd number of them. And p. 339 treats a bridge as producing a face adjacent to itself, so the paper's face language is not confined to simple cycles. We use boundary-walk length throughout.

Condition (d) of Definition 2.1 does not mention bridges, and at a bridge one face lies on both sides of an edge, so the condition is either violated there or vacuous there. Two readings are therefore available in principle, but [1] does choose between them, as we record next.

Convention 2.4 (bridges)**.** Say that a plane graph satisfies *(C2)* if no edge has the same face on both sides, equivalently if it has no bridge. [1] takes (d) to include a face adjacent to itself, so that (C2) follows from Definition 2.1; this is stated explicitly on p. 339, where an alternating plane graph is said to be "always at least 2-edge-connected, since plane graph with edge connectivity 1 contains a face that is adjacent to itself". Under the *weak reading*, (d) constrains only pairs of distinct faces and bridges are permitted.

The proof of Conjecture 10.1 in §4 uses (C2), and we prove in §5 that a putative counterexample must be bridgeless under *either* reading. The result is therefore independent of the choice. We stress that this is a robustness statement and not the repair of a gap: the authors of [1] state the strict reading, and we simply decline to rely on it.

We shall use the following, which is due to [1].

Theorem 2.5 ([1], Theorem 3.2)**.** *In a* $(3,4,5)$*-alternating plane graph,* $v_3 = f_3$*,* $v_4 = f_4$ *and* $v_5 = f_5$*, where* v_i *and* f_i *count vertices of degree* i *and faces of size* i*. Consequently* $v_5 = v_3 - 4$*,* $E = 2V - 2$ *and* $F = V$*.*

2.1. **Rotation systems.** A plane map is encoded by its *rotation system* [3]: for each vertex, its neighbours in clockwise order. Writing α for the involution exchanging the two darts of an edge and σ for the rotation, the faces are the orbits of $\varphi = \sigma^{-1}\alpha$, and the size of a face is the length of its orbit. An edge is a bridge if and only if its two darts lie in the same orbit. Every graph in this paper is presented this way.

3. The per-edge identity

The following double count is the engine of §4. It is the convention behind (9.2) of [1], isolated.

Lemma 3.1. *Let G be a finite plane graph with no isolated vertex, with v vertices, e edges and f faces. For an edge a with ends u, w and face occurrences $F^-(a)$, $F^+(a)$ on its two sides, put*

$$c(a) \;=\; \frac{1}{\deg u} + \frac{1}{\deg w} + \frac{1}{|F^-(a)|} + \frac{1}{|F^+(a)|}.$$

Then $\sum_a c(a) = v + f$. If G is connected, Euler's formula gives

$$\sum_a c(a) \;=\; e + 2, \qquad \textit{equivalently} \qquad \sum_a \big(c(a) - 1\big) \;=\; 2.$$

Proof. A vertex of degree d is an end of exactly d edges and contributes $1/d$ each time, hence 1 in total; summing gives v. By Convention 2.3 a face of size s occurs exactly s times as a side of an edge and contributes $1/s$ each time, hence 1 in total; summing gives f. □

Remark 3.2. Neither count requires facial walks to be simple. Repeated vertices, cut vertices and bridges are all permitted; a bridge simply has $F^-(a) = F^+(a)$ and contributes $2/|F|$ from its face side. This is what makes Lemma 3.1 usable in §5, where bridges are the object of study.

We write $x(a) = c(a) - 1$ for the *excess* of an edge, so that Lemma 3.1 reads $\sum_a x(a) = 2$: in a connected plane graph some edge must have positive excess.

4. Conjecture 10.1

Theorem 4.1. *There is no $2, Y$-alternating plane graph and no $X, 2$-alternating plane graph.*

We prove the two halves separately, then remove the dependence on Convention 2.4 in §5. Both halves are the same inequality with the roles of vertices and faces exchanged; no dual graph is constructed, so neither invokes the 3-edge-connectivity hypothesis of ([1], Lemma 2.2).

4.1. No $2, Y$-alternating plane graph.

Suppose G is one, with degrees taking exactly the two values $d_1 < d_2$.

Adjacent vertices differ in degree and only two degrees occur, so every edge joins a d_1-vertex to a d_2-vertex. In particular G is bipartite, and every edge contributes

$$\frac{1}{d_1} + \frac{1}{d_2} \;\le\; \frac{1}{3} + \frac{1}{4} \;=\; \frac{7}{12}$$

to the vertex side, using $d_1 \ge 3$ from Definition 2.1(a).

Since G is bipartite, every closed walk has even length, so by Convention 2.3 every face has even size, hence size at least 4. By (C2) the two sides of an edge are distinct faces, and by Definition 2.1(d) their sizes differ; being distinct even numbers at least 4, one is at least 4 and the other at least 6. So the face side contributes at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.

Hence $c(a) \le \frac{7}{12} + \frac{5}{12} = 1$ for every edge, so $\sum_a x(a) \le 0$, contradicting Lemma 3.1. □

Remark 4.2. This is (9.5) of [1] with $1/2$ replaced by $1/d_1 \leq 1/3$. The derivation of (9.5) never used the degree 2; §9.1 of [1] runs the chain on the weak class with degrees 2 and k, while §3.2 attacks the strong class with different machinery yielding only $V \geq 25$ and $V \geq 56$ and never bounding f. The two sections were not connected.

4.2. **No $X, 2$-alternating plane graph.** Suppose G is one, with face sizes taking exactly the two values $s_1 < s_2$.

By (C2) the two sides of every edge are distinct faces, and by Definition 2.1(d) they have different sizes; with only two sizes available, every edge separates an s_1-face from an s_2-face and contributes at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ to the face side.

We claim every degree is even. Fix a vertex u of degree d, let its edges in rotation order be $a_1, \dots, a_d$, and let F_i be the face occupying the corner between a_i and a_{i+1}, indices modulo d. The two sides of a_i are the occurrences F_{i-1} and F_i, distinct faces by (C2), so $|F_{i-1}| \neq |F_i|$. With only two sizes the cyclic word $|F_1|, \dots, |F_d|$ alternates, and a cyclically alternating binary word has even length. Hence d is even.

Stating the argument over corners rather than over "the incident faces" is what makes it correct at a cut vertex, where one face may occupy several corners. Two such corners are never separated by a single non-bridge edge, which is all that is needed here; across a bridge, consecutive corners do belong to the same face, and there are no bridges under (C2).

Degrees are at least 3 and even, hence at least 4; adjacent vertices have distinct even degrees, so the vertex side contributes at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. Again $c(a) \leq 1$ for every edge and Lemma 3.1 is contradicted. ☐

Remark 4.3. Euler is used as $v - e + f = 2$; alternating plane graphs are connected by Definition 2.1, and disconnectedness would only strengthen the contradiction.

4.3. **Calibration.** Run on the weak $2, k$ class that [1] settles, the chain excludes exactly $k \geq 12$, reproducing ([1], (9.10)), and does not exclude $k \leq 10$, where that paper exhibits weak $2, k$-alternating plane graphs; $k = 11$ it leaves open, as does [1] without its Lemma 9.4. So the contradiction turns on $d_1 \geq 3$ and not on an over-count, which would also have killed the small-k graphs.

This is emphatically *not* a certification. A spuriously *stronger* face bound such as $5/12 - 1/1000$ reproduces the same threshold exactly, so the calibration cannot detect an invalid strengthening — the dangerous direction. It rules out one failure mode and no others.

5. Removing the dependence on (C2)

Everything in this section is under the weak reading of Convention 2.4: condition (d) of Definition 2.1 constrains only pairs of distinct faces, and bridges are permitted. We show that no $X, 2$-alternating plane graph exists even so, and hence that Theorem 4.1 holds under both readings. Note the scope: this does *not* say that every weak-reading alternating plane graph is bridgeless. It is not — join three copies of a $(3, 4, 5)$-alternating plane graph by two bridges drawn in the exterior, at a degree-5 vertex of each. The joined degrees become $7, 6, 6$, their old neighbours keep degree 3 or 4, the merged exterior face has size at least 13 and so differs from every interior face, and the result is a weak-reading alternating plane graph with bridges. It has neither exactly two degrees nor exactly two face sizes, so it is no counterexample to Theorem 4.1. What is proved here is bridgelessness *within the two-face-size class*. The corresponding statement for $2, Y$ is Proposition 5.5 at the end of the section, proved by the same per-edge count.

The weak reading introduces exactly one new difficulty: a bridge need not force a change of face size, so a vertex on a bridge need not have even degree. The lemma below controls that defect, and the proof then runs as a charge. If a bridge exists, the face carrying it has size at least 8, which makes every bridge contribute excess at most $-\frac{1}{6}$, while the only edges that can have *positive* excess are non-bridge 3–4 edges, each at most $\frac{1}{24}$. The lemma attaches every positive edge to a unique bridge at its degree-3 end, and each bridge receives at most two. The bridges' deficit then dominates every positive excess, contradicting $\sum_a x(a) = 2$. If there is no bridge, the even-degree argument of §4.2 applies directly.

Lemma 5.1 (parity)**.** *Let G be a plane graph whose faces admit a colouring by two colours that is proper on every pair of distinct adjacent faces. Then for every vertex v,*

$$\deg v \;-\; \bigl|\{\textit{bridges at } v\}\bigr| \quad \textit{is even.}$$

Proof. Walk the darts at v in rotation order and let $F_1, \ldots, F_d$ be the corner faces as in §4.2, so that F_{i-1} and F_i are the two sides of the edge a_i. These two occurrences are of the same face precisely when a_i is a bridge. Hence the corner colour changes across each non-bridge edge at v and is unchanged across each bridge at v. The walk closes up, so the number of changes is even, and that number is $\deg v$ minus the number of bridges at v. □

Lemma 5.1 is the bridge-tolerant replacement for the classical Eulerian criterion, and it is strictly weaker than that criterion rather than a restatement of it: a triangle with a pendant edge has its two faces properly coloured once self-adjacency across the bridge is ignored, yet is not Eulerian. With no bridges the lemma gives even degree at every vertex — the direction of the usual Eulerian criterion needed here, and the only one it supplies. In an $X, 2$-alternating plane graph the face size supplies the colouring, so the lemma applies.

Corollary 5.2. *In an $X, 2$-alternating plane graph under the weak reading, a vertex lying on no bridge has even degree, hence degree at least* 4.

Theorem 5.3. *Under the weak reading there is no $X, 2$-alternating plane graph. In particular no such graph has a bridge.*

Proof. Let G be one, with face sizes $s_1 < s_2$, and let B be its number of bridges. Suppose first $B \geq 1$.

Step 0: $s_2 \geq 8$. Let $a = uw$ be a bridge, and let A and C be the components of $G - a$ containing u and w. Then $\deg_A u = \deg u - 1 \geq 2$ and likewise $\deg_C w \geq 2$, so both have at least three vertices. Let W_A be the facial boundary walk of A through the corner exposed by deleting a. Its length is not 1, which needs a loop. Nor 2: that is either a digon of parallel edges, excluded by simplicity, or a single edge traversed in both directions, and for those two darts to form the whole facial boundary both endpoints of that edge have degree 1, forcing $A = K_2$ against $\deg_A u \geq 2$. So $|W_A| \geq 3$, and likewise $|W_C| \geq 3$. The face F of G incident with both sides of a is traversed as W_A, then a, then W_C, then a in the opposite direction, so by Convention 2.3

$$|F| \;=\; |W_A| + |W_C| + 2 \;\geq\; 3 + 3 + 2 \;=\; 8.$$

As $|F| \in \{s_1, s_2\}$ and $s_1 < s_2$, we get $s_2 \geq 8$.

Step 1: only an edge at a vertex of degree 3 *can have positive excess.* Recall $s_1 \geq 3$ and, by Step 0, $s_2 \geq 8$; degrees at the two ends of an edge are distinct by Definition 2.1(c). Three cases:

- a a bridge: both sides carry F, so $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6}$ and $x(a) \leq -\frac{1}{6}$;
- a not a bridge, both ends of degree at least 4: $c(a) \leq \frac{1}{4} + \frac{1}{5} + \frac{1}{3} + \frac{1}{8} = \frac{109}{120}$ and $x(a) \leq -\frac{11}{120}$;
- a not a bridge with an end of degree 3: $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{3} + \frac{1}{8} = \frac{25}{24}$ and $x(a) \leq \frac{1}{24}$.

Only the third class can be positive. Note in particular that a bridge endpoint of degree 5 cannot make an edge positive.

Step 2: each bridge carries at most two positive edges. First, a positive edge is a 3–4 edge. Its face side is at most $\frac{1}{3} + \frac{1}{8} = \frac{11}{24}$, so its vertex side must exceed $\frac{13}{24}$; among pairs of distinct degrees at least 3, only $\frac{1}{3} + \frac{1}{4} = \frac{14}{24}$ does, since already $\frac{1}{3} + \frac{1}{5} = \frac{12.8}{24}$ falls short.

Let a be an edge of positive excess and let u be its end of degree 3, which is unique since adjacent vertices have different degrees. By Lemma 5.1 the number of bridges at u has the same parity as $\deg u = 3$, so it is 1 or 3; it is not 3, since a itself is a non-bridge edge at u. So u lies on exactly one bridge. Write $b(u)$ for it and *charge* a to $b(u)$: every positive edge thus has one, and only one, destination.

The qualification matters: a vertex of degree 3 need not carry two non-bridge edges, since one with three bridges carries none. What is proved, and what is needed, is that a vertex of degree 3 *incident with a non-bridge edge* lies on exactly one bridge and two non-bridge edges.

Now fix a bridge b and count the positive edges charged to it. Each such edge has its degree-3 end at a vertex whose unique bridge is b, so that vertex is an end of b. The two ends of b have different degrees by Definition 2.1(c), so at most one of them has degree 3; call it u if it exists. Of the three edges at u, one is b itself, leaving exactly two that could be positive and charged to b. So each bridge receives at most two charges, and writing P for the number of positive edges, $P \leq 2B$.

Step 3. Partition the edges into the P positive ones, the B bridges, and the rest. By Steps 1 and 2 their excesses are at most $\frac{1}{24}$, $-\frac{1}{6}$ and 0 respectively, so by Lemma 3.1,

$$2 \;=\; \sum_a x(a) \;\leq\; \frac{P}{24} - \frac{B}{6} \;\leq\; \frac{2B}{24} - \frac{B}{6} \;=\; -\frac{B}{12} \;<\; 0,$$

a contradiction.

Finally suppose $B = 0$. By Corollary 5.2 every degree is even and at least 4, so the vertex side of every edge is at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$; the face side is at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$; hence $x(a) \leq 0$ for every edge and $\sum_a x(a) = 2$ is again contradicted. □

Remark 5.4 (how much room the argument has)**.** The tether of Step 2 is the substance: treating the positive edges as unconstrained gives only a lower bound on B and closes nothing. It does not have to be sharp. Ignoring the degree condition on a bridge's two ends gives only $P \leq 4B$, and even then the bound of Step 3 is 0, which still contradicts 2; the sharp tether merely supplies the margin $-B/12$. That margin also means Step 0 is not needed at full strength: with $s_2 \geq 7$ the per-bridge total is $-\frac{1}{84}$, still negative. Some lower bound on the bridge face is nevertheless indispensable — at $s_2 \geq 6$ the total is $+\frac{1}{12}$ — and at $P \leq 5B$ the argument fails outright.

Proposition 5.5. *Under the weak reading there is no* $2, Y$*-alternating plane graph either.*

Proof. Step 0 of Theorem 5.3 uses only minimum degree 3 and simplicity, so it applies here: a bridge's face has size at least 8. As in §4 the graph is bipartite, since two degrees and Definition 2.1(c) force every edge to join a d_1-vertex to a d_2-vertex; hence every closed walk has even length, a facial walk traversing a bridge twice included, and every face has even size at least 4. Then a bridge has

$$c(a) \;\leq\; \tfrac{1}{3} + \tfrac{1}{4} + \tfrac{2}{8} \;=\; \tfrac{5}{6},$$

and a non-bridge edge, whose two sides are distinct faces of distinct even sizes at least 4 and 6, has $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} = 1$. So $x(a) \leq 0$ for every edge, contradicting $\sum_a x(a) = 2$. □

6. The periodic capping lemma

Conjecture 10.2 asks for a $(3, 4, 5)$-alternating plane graph on n vertices for every $n \geq 20$. Twenty-six orders were open. We supply certificates for all of them (§9) and, more usefully, prove that a single certificate generates an infinite family, so that the result does not rest on a finite list.

6.1. **The cover.** The certificates are capped unrollings of a periodic quotient of the torus. The relevant structure is a bi-infinite *strip*, built from identical *copies*, each contributing three vertices and six edges, together with two *caps* that close it off at the ends. Writing TARGET_n for a certificate of order n, its alignment against the cover exhibits a maximal run of copies whose rotations are translates of one another; we call this run the *deep block* D.

Theorem 6.1 (periodic capping lemma)**.** *Fix an order n whose alignment against the cover has a deep block D of at least five copies, and let* cut *be a copy at distance at least two from each end of D. Then for every integer $d \geq -(|D|-1)$ the rotation system* $\mathrm{splice}(n, d)$*, obtained from* TARGET_n *by inserting d further translates of the copy* cut *(or deleting $-d$ of them when $d < 0$), is a $(3, 4, 5)$-alternating plane graph on $n + 3d$ vertices.*

Proof. Write $S = \mathrm{splice}(n, d)$. Every vertex of S receives its rotation in one of four ways: cap vertices verbatim, with strip neighbours above the cut shifted by d; strip copies at or below the cut verbatim; strip copies above $\mathrm{cut} + d$ verbatim from copy $c - d$; and the d fresh copies from the cut copy's rotation, translated.

The argument is local. A facial walk is traced by $\varphi = \sigma^{-1}\alpha$, which reads only the edge partner and the rotation predecessor at each vertex it visits.

The span bound is taken on TARGET_n, not on S. This matters: a bound measured on S could not be used here without circularity, since it is the faces of S that are in question. In TARGET_n — a verified $(3, 4, 5)$-alternating plane graph — every face has at most five darts, and the cover's largest edge offset is two. A closed sequence of at most five steps, each of size at most two, has span at most four: reaching level 5 needs three ascents, leaving two steps to descend five, which is impossible. So *every face of* TARGET_n *lies inside five consecutive copies.* Both inputs are finite checks on the one certificate (Remark 6.2).

Four is the honest bound, and it is not sharp: the largest span actually attained, over every spliceable order at $d = 1, 2, 3, 5$, is two. But sharpness is not needed, because five copies is exactly what the hypothesis supplies. The fresh copies lie in $(\mathrm{cut}, \mathrm{cut} + d]$ and are translates of cut; since cut is at distance at least two from each end of D, the copies $\mathrm{cut} - 2, \ldots, \mathrm{cut} + 2$ all lie in D and are mutual translates. Hence every copy in $[\mathrm{cut} - 2, \mathrm{cut} + d + 2]$ is a translate of cut, and every five-copy window meeting the fresh block is a translate of $[\mathrm{cut} - 2, \mathrm{cut} + 2]$; away from that range S agrees with TARGET_n verbatim.

Now trace φ in S from any dart. Its counterpart trace in TARGET_n closes within five steps inside a five-copy window, and S is a translate of TARGET_n on that window, so each step of the trace in S reproduces the corresponding step and the walk closes at the same length. Hence *every face of S is a translated face of* TARGET_n.

Each condition of Definition 2.1 is now inherited, because each is local: face sizes and vertex degrees lie in $\{3, 4, 5\}$; adjacent vertices have unequal degrees and adjacent faces unequal sizes; facial walks are simple; and a loop or parallel edge would have to sit inside a two-copy window, where S and TARGET_n agree. Connectivity is inherited because each copy meets the next and every cap vertex reaches the strip as before.

Finally S is spherical. A copy contributes three vertices, six edges and — since every face is a translated face — three faces, and $3 - 6 + 3 = 0$, so $V - E + F$ is unchanged from TARGET_n, where it is 2. □

Remark 6.2 (what is machine-checked)**.** Three facts in the proof are not bookkeeping and are discharged by computation on the single certificate TARGET_n: that every face has at most five darts, that no strip edge joins copies more than two apart, and that the deep block really is periodic, i.e. its rotations are translates of one another. All three are finite checks on one graph, and none is a statement about S. Everything else is the local argument above.

6.2. **What the family reaches.** The floor $-(|D| - 1)$ is where deletion would consume a copy that is not deep, that is, where the two cap collars meet; each collar is two copies wide. Splicing changes the order by $3d$, so a certificate reaches one residue class modulo 3. The floors are:

n mod 3	floor	family
0	48	$48, 51, 54, 57, \dots$
1	52	$52, 55, 58, 61, \dots$
2	50	$50, 53, 56, 59, \dots$

Corollary 6.3. *Conjecture 10.2 holds: there is a* $(3, 4, 5)$*-alternating plane graph on* n *vertices for every* $n \geq 20$*.*

Proof. Orders below 46 are the published witnesses of [1], decoded from the author-hosted corpus [7], together with the Section-8 closures of §7. Orders 37 and 38 are covered by the heuristic search of [1]; the disk surgery of this paper supplies independently checked witnesses there, needed for §7 rather than to fill a gap in Conjecture 10.2. Orders 46 through 56, 67 through 74, 88 through 92, and 109, 110 carry the certificates of this paper. Every remaining order is reached by Theorem 6.1 from a certificate in its residue class modulo 3, with the floors above. ☐

Remark 6.4 (the deletion direction)**.** Theorem 6.1 is stated for every $d \geq -(|D| - 1)$, and the floors in the table are reached by deletion rather than insertion. The locality argument is written above for insertion, where the fresh copies are interior to D; for $d < 0$ it reads the same way with the four cases collapsing, save at the floor itself, where a three-copy window meets both cap collars and "every window is a translated window" is no longer literally true. The floor orders are therefore also carried by their own verified certificates.

The reach must be measured against the hypothesis $|D| \geq 5$, not against the list of certificates. Of the twenty spliceable orders, five — 48, 51, 53, 54 and 56 — have deep blocks of only $1, 2, 2, 3$ and 3 copies and are therefore *not* eligible seeds. The eligible ones begin at 67, 68 and 69, one per residue class, so insertion reaches nothing below 70, and **all ten orders** 57–66 **lie outside Theorem 6.1**. None of them carries a stored certificate either. What supports them is the verified computation of `test_pumping_splice.py`, which builds successive splices from 90, 109 and 110 and runs each through the decision procedures.

Two earlier versions of this remark got the set wrong — first $57, 58, 59, 61, 63$, then $58, 61, 64$ — the second because it was computed from the spliceable orders rather than from the theorem's hypothesis. Being computed rather than asserted does not help if the predicate is the wrong one. It is now derived from $|D| \geq 5$ directly (`test_conjecture_coverage.py`). Stating and proving the deletion direction separately, with its own cut rule, is left open.

This is an independent construction of ([1], Theorem 8.1), which gives $n \geq 111$, and of twenty-three of the twenty-six open orders, from one certificate per residue class. Orders 46, 47 and 49 lie outside the family and rest on their certificates alone.

Consequently the certificates and Theorem 6.1 together settle *every order* $n \geq 46$ without reference to [1], subject to Remark 6.4 at the five orders noted there; Theorem 8.1 is not needed. Orders 20 through 45 are inherited from that paper and are not re-established here.

Remark 6.5 (why this is not twenty-six more examples)**.** The splice reproduces certificates it did not start from. TARGET_{110} shortened by twenty periods is TARGET_{50} up to isomorphism of rotation systems, and so is TARGET_{92} shortened by fourteen; TARGET_{90} shortened by fourteen and TARGET_{72} shortened by eight both give TARGET_{48}. Two independently found certificates decompose into the same pair of caps and the same period, which is what shows the cap extraction to be canonical rather than an artefact of one alignment.

7. Conjecture 10.3

Theorem 7.1. *There is a* 3*-connected alternating plane graph on* n *vertices for*

$$n \in \{17\} \cup [19, 56] \cup [67, 74] \cup [88, 92] \cup \{109, 110\},$$

54 *orders in all, each certified by an explicit witness. (Order* 17 *is below the range of Conjecture 10.3; it is included because it is the smallest order at which any alternating plane graph exists, and its witness is* 3*-connected.)*

We state Theorem 7.1 in this bounded form deliberately. Conjecture 10.3 asks for every $n \geq 19$, and we do not prove that: the remaining orders 57–66, 75–87, 93–108 and $n \geq 111$ are carried only by the spliced family, whose 3-connectivity induction we have been unable to establish (Remark 7.3). An earlier version of this paper stated the full conjecture as a theorem; that was withdrawn on 2026-09-05.

The witnesses come from six sources, and no order depends on more than one of them being correct.

orders	source
19	all five alternating plane graphs on 19 vertices
$17, 20, 26$–$36, 42$	published witnesses, decoded and re-verified
$21, 22, 23, 25$	published search seeds
21–24, 39–45	Section-8 closures, rebuilt here
$37, 38$	disk surgery
46–56, 67–74, 88–92, 109, 110	the certificates of §6
withdrawn 2026-09-05: 48 and every $n \geq 50$, the spliced family	

Three orders deserve comment.

Order 19 is the one order the settled Conjecture 10.2 cannot reach: [1] reports no $(3, 4, 5)$-alternating plane graph on 18 or 19 vertices, so a witness must be a general alternating plane graph in the sense of Definition 2.1. There are exactly five on 19 vertices [6], and all five are 3-connected. A single one would have sufficed.

Orders 37 *and* 38 are out of reach of the Section-8 arithmetic $18a + 19b + 20c + 21d + 3$, because 34 and 35 are not sums of $18, 19, 20, 21$. They are *not* outside [1] altogether — its heuristic search reaches every order from 20 to 42 — but its *construction* misses them, so a witness built here is the first that is checkable from a stored rotation system. They are obtained by disk surgery: excise the star of a vertex, an edge or a face from a stored $(3,4,5)$-alternating plane graph, let the boundary degrees float, and refill the disk.

The infinite tail is not established. An earlier version argued that the spliced family is 3-connected at every order it produces, by the same locality that carries Theorem 6.1. That induction does not go through; it was withdrawn on 2026-09-05 and the defects are set out in Remark 7.3. Orders 57–66, 75–87, 93–108 and every $n \geq 111$ are therefore open here.

7.1. **A shortcut that does not exist.** If every $(3,4,5)$-alternating plane graph were 3-connected, then Corollary 6.3 would give Theorem 7.1 for all $n \geq 20$ in one step.

Proposition 7.2. *There is a* $(3,4,5)$*-alternating plane graph on* 46 *vertices with a separating pair. In particular* $(3,4,5)$*-alternating plane graphs need not be* 3*-connected.*

The graph is supplied as a rotation system in the artifact; both verifiers accept it, its unique separating pair is exhibited, and removing that pair leaves two components of 22 vertices each. Nothing in Theorem 7.1 depends on the proposition — every witness is checked individually rather than inferred from a class-wide claim — but it is why the connectivity of the spliced family has to be proved rather than assumed.

Remark 7.3 (the infinite tail is not proved)**.** Adversarial review on 2026-09-05 found the induction carrying 3-connectivity through the spliced family to be unestablished, and we record that here rather than in a companion file. Its key step shows a freshly spliced copy is adjacent to the copies on both sides, and concludes that connectivity is preserved; what is required is that $S(n,d) - \{u,v\}$ is connected, with the pair already removed, which does not follow. A second step — that the splice point can always be placed clear of a pair occupying up to four consecutive copies, inside a deep block of at least five — is asserted rather than checked.

Consequently: verified witnesses exist at 17, 19–56, 67–74, 88–92 and 109–110, and none of them depends on the induction. Orders 57–66, 75–87, 93–108 and every $n \geq 111$ do. [1] states in its concluding remarks that the $(3,4,5)$-alternating plane graphs of its Section 8 are 3-connected, which would cover exactly those gaps, but that is an assertion rather than a theorem there. We therefore claim Conjecture 10.3 settled where witnesses exist and open pending proof above that.

8. The asymptotic degree distribution

The fourth problem of ([1], §10, p. 362) is not a counting question but a question about the shape of a typical graph. Writing $r = v_3$, Theorem 2.5 confines v_4 to $[r-5, \frac{3}{2}r-5]$, so the ratio v_4/v_3 lies asymptotically in $[1, 1.5]$. The authors ask whether there is a density function on that interval giving the asymptotic fractions of $(3,4,5)$-alternating plane graphs for large n, and if so what it looks like.

We have nothing to offer on it. What we can add is a caution, from the only data this paper produces.

All twenty-six certificates satisfy $v_5 = v_3 - 4$, as they must, and all lie inside the interval. But they lie at its very bottom: v_4/v_3 ranges over $[1.0000, 1.1250]$ across the twenty-six, and drifts *downwards* with n, attaining exactly 1 at orders 50, 56, 74, 92 and 110.

That is a selection effect, not a measurement. The certificates come from one construction — a capped unrolling of a periodic strip — and Theorem 6.1 extends it by inserting copies of a single period, so every member of the family inherits that period's degree profile. A construction emitting one graph per order reports on the construction, not on the distribution over all graphs of that order. The clustering at $v_4/v_3 = 1$ is precisely what a periodic family looks like and precisely what a sample would not.

The methods of this paper cannot reach the question, for a structural reason. Everything proved above is an existence statement: one graph per order, or a class shown empty. A density function requires a measure over all graphs of order n, and nothing here enumerates. The class is moreover rigid in a way that obstructs sampling: a sweep over every two-edge switch of all twenty-six certificates finds no switch leaving a valid $(3, 4, 5)$-alternating plane graph, so the one-switch neighbourhood of each is empty and local moves cannot walk the space.

Three routes seem plausible. An exhaustive census at small orders, with `plantri` [5] or an equivalent generator restricted to the degree and face-size profile, would give the first honest histogram of v_4/v_3; it is a means to a conjecture rather than to a theorem, and it is where we would start. A structure theorem giving *every* $(3, 4, 5)$-alternating plane graph a cap–strip–cap decomposition with a bounded interface alphabet would make the count a regular language and the density a transfer-matrix computation; the obstacle is the word *every*, since we have a construction and not a classification. Failing either, a bijection with an already-counted family of plane maps would serve, and the identities of Theorem 2.5 are the natural hook.

9. The artifact

Every graph asserted in this paper is supplied as an explicit rotation system in the accompanying artifact, in a documented JSON format with a dependency-free reference reader. No file records a claim that a graph *is* an alternating plane graph: degrees, faces, face sizes, bridges, connectivity and both alternation conditions are recomputed on every run by four separately written decision procedures. Certificates can also be exported to `planar_code` for checking by `plantri` [5] or House of Graphs [6].

The artifact runs 1308 checks. The four decision procedures use the Python standard library alone; the automated suite that drives them additionally requires `pytest`, and nothing else. Two classes of check are load-bearing: that each certificate is accepted by every verifier of its class — the order-19 witnesses are general alternating plane graphs and the three $(3, 4, 5)$ procedures correctly reject them — and that each source in Theorem 7.1 is necessary, in the sense that removing it reopens orders. Most gates are paired with a control that must fail, so that a check cannot pass by being vacuous. Not all: `test_verifier_mutations.py` lists 17 sites with no negative control, and in September 2026 three coverage gates were found to be iterating an empty set. Both are recorded in the artifact rather than corrected in silence. Of the 17, six are the Theorem 3.2 identities, two are the facial-walk restriction the definition does not impose, and the two alternation clauses are each controlled in the *other* verifier, whose checks run in a different order; every certificate is run through both.

The two computations that the proofs genuinely rest on are the finite hypotheses of Remark 6.2.

Data availability. The artifact — every certificate, all four decision procedures, the drawing tool (a straight-line embedding after Tutte [4]) and the export tools, and the complete test suite — is archived at [2], under the MIT licence. The concept DOI 10.5281/zenodo.22269200 always resolves to the current version; this paper describes version 1.0.4, 10.5281/zenodo.22666407. Reproduce with `make deps && make verify`. No third-party *files* are redistributed: no licence statement was found at the source of the published corpus [7], so each of its graphs is re-expressed in the artifact's own format with the SHA-256 of every original retained for comparison. The graphs themselves are of course the corpus's; re-encoding changes the bytes, not the provenance, and the artifact's licence is not asserted over them.

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